Cristian Mesquida

Postdoctoral Researcher

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Employment

Postdoctoral researcher — *Eindhoven, The Netherlands*

Eindhoven University of Technology — *2025-Present*

- Contributed to the development and validation of Metacheck, an R package for improving research quality and extracting data from published articles

PhD candidate — *Eindhoven, The Netherlands*

Eindhoven University of Technology — *2021-2025*

- My PhD thesis examined the factors hindering replicability and reproducibility in sports and exercise science

Education

PhD — *Eindhoven, The Netherlands*

Eindhoven University of Technology — *2021-2025*

- My PhD thesis focused on the factors that hinder replicability and reproducibility of research in sports and exercise science

MSc. Integrative Physiology — *Barcelona, Spain*

University of Barcelona — *2017-2019*

- My master thesis focused on the effect of two training protocols on maxymal oxygen uptake in highly-trained cyclists

MSc. Research in Sport and Physical Activity — *Granada, Spain*

University of Granada — *2015-2017*

- My master thesis focused on the performance determinants of swimming sprint performance

BSc. Sports and Exercise Science — *Vic, Spain*

University of Vic — *2009-2013*

- My bachelor end project focused on developing a training programe to reduce the injury risk of hamstrings in football players

Publications

1. Lakens, D., Mesquida, C., Xavier-Quintais, G., Rasti, S., Toffalini, E., & Altoè, G. (2026). Rethinking Type S and Type M Errors. *Advances in Methods and Practices in Psychological Science*, *9*(2), 25152459261432530. https://doi.org/10.1177/25152459261432530
2. Ditroilo, M., Mesquida, C., Abt, G., & Lakens, D. (2025). Exploratory research in sport and exercise science: Perceptions, challenges, and recommendations. *Journal of Sports Sciences*, 1–13. https://doi.org/10.1080/02640414.2025.2486871
3. Murphy, J., Caldwell, A. R., Mesquida, C., Ladell, A. J. M., Encarnación-Martínez, A., Tual, A., Denys, A., Cameron, B., Van Hooren, B., Parr, B., DeLucia, B., Mason, B. R. J., Clark, B., Egan, B., Brown, C., Ade, C., Sforza, C., Taber, C. B., Kirk, C., … Warne, J. P. (2025). Estimating the Replicability of Sports and Exercise Science Research. *Sports Medicine*. https://doi.org/10.1007/s40279-025-02201-w
4. Lakens, D., Mesquida, C., Rasti, S., & Ditroilo, M. (2024). The benefits of preregistration and Registered Reports. *Evidence-Based Toxicology*, *2*(1), 2376046. https://doi.org/10.1080/2833373X.2024.2376046
5. Mesquida, C., & Lakens, D. (2024). Is the effect large enough to matter? Why exercise physiologists should interpret effect sizes meaningfully: A reply to Williams et al. (2023). *The Journal of Physiology*, *602*(1), 241–242. https://doi.org/10.1113/JP285901
6. Holgado, D., Mesquida, C., & Román-Caballero, R. (2023). Assessing the Evidential Value of Mental Fatigue and Exercise Research. *Sports Medicine*, *53*(12), 2293–2307. https://doi.org/10.1007/s40279-023-01926-w
7. Mesquida, C., Murphy, J., Lakens, D., & Warne, J. (2023). Publication bias, statistical power and reporting practices in the Journal of Sports Sciences: Potential barriers to replicability. *Journal of Sports Sciences*, *41*(16), 1507–1517. https://doi.org/10.1080/02640414.2023.2269357

8. Murphy, J., Mesquida, C., & Warne, J. (2023). A Survey on the Attitudes Towards and Perception of Reproducibility and Replicability in Sports and Exercise Science. *Communications in Kinesiology*, *1*(5). https://doi.org/10.51224/cik.2023.53
9. Bossi, A. H., Mesquida, C., Hopker, J., & Rønnestad, B. R. (2023). Adding Intermittent Vibration to Varied-intensity Work Intervals: No Extra Benefit. *International Journal of Sports Medicine*, *44*(2), 126–132. https://doi.org/10.1055/a-1812-7600
10. Mesquida, C., Murphy, J., Lakens, D., & Warne, J. (2022). Replication concerns in sports and exercise science: A narrative review of selected methodological issues in the field. *Royal Society Open Science*, *9*(12), 220946. https://doi.org/10.1098/rsos.220946
11. Murphy, J., Mesquida, C., Caldwell, A. R., Earp, B. D., & Warne, J. P. (2022). Proposal of a Selection Protocol for Replication of Studies in Sports and Exercise Science. *Sports Medicine*. https://doi.org/10.1007/s40279-022-01749-1
12. Etxebarria, N., Wright, J., Jeacocke, H., Mesquida, C., & Pyne, D. B. (2021). Running Your Best Triathlon Race. *International Journal of Sports Physiology and Performance*, *16*(5), 744–747. https://doi.org/10.1123/ijspp.2020-0838
13. Bossi, A. H., Mesquida, C., Passfield, L., Rønnestad, B. R., & Hopker, J. G. (2020). Optimizing Interval Training Through Power-Output Variation Within the Work Intervals. *International Journal of Sports Physiology and Performance*, *15*(7), 982–989. https://doi.org/10.1123/ijspp.2019-0260

Preprints

1. Mesquida, C., Warne, J., & Lakens, D. (2025). *A scoping review of the transparency, reporting practices and methodological rigor of meta-analyses published in Sports Medicine*. SportRxiv. https://doi.org/10.51224/SRXIV.576
2. Mesquida, C., Murphy, J., Warne, J., & Lakens, D. (2025). *Prevalence, reporting practices, and methodological quality of a priori power analyses in sports and exercise science research*. SportRxiv. https://doi.org/10.51224/SRXIV.575
3. Mesquida, C., Warne, J., & Lakens, D. (2025). *What is your hypothesis? On the importance of knowing your hypothesis before conducting a hypothesis test*. SportRxiv. https://doi.org/10.51224/SRXIV.577
4. Mesquida, C., Murphy, J., Warne, J., & Lakens, D. (2025). *On the replicability of sports and exercise science research: Assessing the prevalence of publication bias and studies with underpowered designs by a z-curve analysis*. SportRxiv. https://doi.org/10.51224/SRXIV.534

Papers in preparation

1. DeBruine, L., Mesquida, C., & Lakens, D. (2026). *Introducing Metacheck as an Automated Tool for Improving Research Practices in Psychology*.
2. Mesquida, C., DeBruine, L., & Lakens, D. (2026). *Improving the Transparency of Power Analyses Through Standardized Reporting*.
3. Lakens, M., Daniël, & DeBruine, L. (2026). *Best Practices for Machine Readable Data and Code Sharing in Psychology*.
4. Nolte, S., Muprhy, J., Mesquida, C., & Rein, R. (2027). *Computational Reproducibility in Sports Science: A Registered Report Reanalysing Private Raw Data*.

Teaching Experience

My teaching portofolio can be found **here**

Teacher of Advanced Research Methods and Research Ethics

Eindhoven University of Technology *2025-Present*

- Taught p-values and Bayes factors, power analysis, and equivalence testing

Teacher of Behaveoral Research Methods 2: Dealing with data

Eindhoven University of Technology *2025-Present*

- Taught multiple regression, variable selection and power analysis

Teacher of Metascience

Eindhoven University of Technology *2024-Present*

- Taught introduction to meta-analysis and bias-detection tests

Teaching assistant of Advanced Research Methods and Research Ethics

Eindhoven University of Technology *2024*

- Assisted students during tutorial sessions with course assignments

Teaching assistant of Advanced Data Analysis

Eindhoven University of Technology *2024*

- Taught R Markdown for reproducible reporting and data simulation for statistical analysis

Teaching Assistant of Applied Exercise Physiology

Technological University of Dublin *2021*

- Taught anatomy

Supervision

Supervisor of Master End Project

Eindhoven University of Technology *2026*

- Co-supervised a project creating an automated tool that categorizs repository files based on the file name, index detected variables and their labels, and reconstruct the repository into an established standard

Supervisor of Master End Project

Eindhoven University of Technology *2025*

- Co-supervised a project using automated search extraction to extract and attempted hypotheses using large language models

Supervisor of Bachelor End Project

Eindhoven University of Technology *2025*

- Co-supervised a meta-analysis project examining the impact of school sanitation on students' academic performance

Supervisor of Master End Project

Eindhoven University of Technology *2024*

- Co-supervised the development of a tool to identify incorrect use of the standard error of measurement instead of the standard deviation in effect size calculations for meta-analyses

Qualifications

Start to Supervise *The Netherlands*

Eindhoven University of Technology *2026*

- Enabling Course for main supervisors

National Triathlon Coach *Spain*

Spanish Triahtlon Federation *2020*

Talks

Lunch Talk

German Sport University Cologne *2026*

- Discussed the main quality issues in power analyses reported in sports and exercise science research

FORRT AI in Metascience

FORRT *2026*

- Discussed the lack of transparency in power analysis reporting in psychology and how Metacheck can help improve reporting practices

ONH Metascience Symposium

Oslo Nye Høyskole *2026*

- Presented findings on the quality of power analyses for sample size justification in sports and exercise science

Transparency, Technology and AI in Peer Review

Eindhoven University of Technology *2026*

- Presented Metacheck, an automated tool for assessing research transparency

Metascience Series

Charles Sturt University *2025*

- Gave a talk on research waste in sports and exercise science

Workshop on Metacheck

Open Research Summer School at King's College London *2025*

- Delivered a workshop on Metacheck, including a presentation and hands-on session

Danish Reproducibility Network meetup

Danish Reproducibility Network *2025*

- Gave a talk on the benefits of preregistration and Registered Reports in hypothesis-testing research

Lunch meetup for the Open Science Community

Eindhoven University of Technology *2025*

- Gave a talk on the benefits of preregistration and Registered Reports in hypothesis-testing research

NLRN Pre-Symposium for Young Meta-Scientists

University Medical Center Groningen *2024*

- Presented findings on publication bias and statistical power in sports and exercise science

Paul Meehl Graduate School PhD Day Conference

Eindhoven University of Technology *2024*

- Presented findings on the reporting and reproducibility ofa priori power analyses in sports and exercise science

Workshop on endurance performance

Setanta College *2021-2023*

- Conducted a workshop on the physiological determinants of endurance and team-sport performance, including lactate-threshold testing

Workshop on 1080 Sprint resistance device

Kumming Hiageng Physical Training Base *2019*

- Conducted a workshop on using the 1080 Sprint to assess swimmers' power–velocity profiles and drag

Grants

Replication studies

Dutch Research Council *2026*

- Applied as a co-applicant. The funded project aims to assess the quality and reproducibility of power analyses in large-scale replication projects. (Project details at https://doi.org/10.61686/FSWVF35053)

IE&IS ESA Education Innovation Grant

Eindhoven University of Technology *2024*

- Applied as a co-applicant. The funded project aims to develop an Metacheck modules designed to assist master's students and supervisors in performing automatic checks to enforce best research practices

Conference organization

Transparency, Technology and AI in peer review Conference *The Netherlands*

Eindhoven University of Technology *2026*

- I co-organized a one-day conference that brought together different stakeholders to discuss the role of transparency, technology, and AI in peer review (https://meta-eindhoven.github.io/events/)

Skills

Advanced R

- Data wrangling, statistical modelling, simulation, Quarto

Basic Python

- I am currently enrolled in a course to learn Python

Medium Git/Github

- Version control, collaborative workflows, project management

Statistical analysis

- Regression, mixed-effects models, meta-analysis, power analysis

Data management

- Data organisation, documentation, FAIR data and code